Sustainable AI Assessment Platform
for Algorithmic Problem Solving

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The Context

- Algorithmic answers require **descriptive explanation** of logic, steps, and complexity.
- Evaluation is currently **manual**, repetitive, and prone to fatigue.

Why This Matters

- **Scalability:** Increasing class strengths make detailed feedback impossible.
- **Subjectivity:** Maintaining grading consistency across different evaluators is challenging.

"A need for efficient, transparent, and scalable assessment."

Problem Definition

Develop a **sustainable and privacy-preserving AI system** to automatically evaluate descriptive algorithmic answer sheets using structured, rubric-based scoring.

Scope: Focus on handwritten descriptive solutions using **OCR** and **Small Language Models (SLMs)**.

Constraint: The system must run **locally** (offline) to ensure data privacy and sustainability.

Impact: Faster grading cycles and consistent feedback for students.

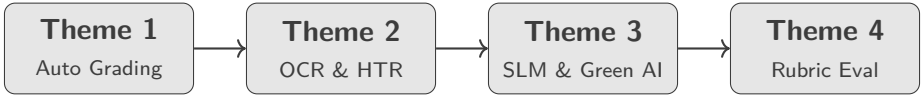
Limitations of Existing Systems

- **Format:** Mostly limited to MCQs or executable code test-cases.
- **Privacy:** Heavily dependent on cloud APIs (OpenAI/Gemini), risking student data.
- **Cost:** High token costs for large-scale deployment.

Our Proposed Edge

- **Offline Deployment:** No data leaves the institution.
- **Green AI:** Uses lightweight models (SLMs) instead of massive LLMs.
- **Explainability:** Scores are tied to a specific rubric, not a "black box."

We conducted a comprehensive review of **40+ papers** (2024-2025) categorized into four strategic themes:



The following slides detail the most critical papers from each theme.

Focus Area	Author (Paper)	Method	Context	Dataset	Critical Insight
GenAI Scoring	Pecuchova et al. (2025)	Semantic-level analysis using GPT-4 & Claude 3	Grading Open-Ended Short Answers	Course-Specific Student Responses	LLMs match human grading in higher-order thinking but require **rubric alignment**.
Explainable AI	Thakur et al. (2025)	TrueGradeAI (RAG + Bias Mitigation)	Secure Digital Examinations	Simulated Exam Scripts	**Explainability** (Why a grade was given) is essential for user trust and auditability.
Review	Emirtekin (2025)	PRISMA Systematic Lit. Review	AI in Education 4.0	ERIC / Scopus Database	Highlights unresolved issues in **fairness**, data privacy, and algorithmic bias.
Code Eval	Bernik et al. (2024)	LLM-based Static Analysis	Introductory Programming (Python)	Private Student Submissions	AI reduces time for executable code but often fails on **conceptual pseudocode**.
Formative	Mendonça et al.	Automated Feedback Scoring	Science Education Scaffolding	Climate Change Module Logs	**Prompt sensitivity** significantly alters grading accuracy in formative tasks.
Feedback	Patil et al.	Integrated Actionable Feedback	University Classrooms	Internal Course Data	Combining grading with **actionable feedback** improves learning outcomes.
Simulation	Discover AI	Simulation-Based Assessment	Medical Performance Tasks	Clinical Sim. Data	AI works well for performance tasks applicable to **step-wise logic**.

Focus Area	Author (Paper)	Method	Context	Dataset	Critical Insight
Layout Analysis	Microsoft (2022)	LayoutLMv3 (Multimodal Transformer)	Document Structure & Form Parsing	FUNSD, CORD	Preserving **indentation** is vital for parsing algorithmic logic in code.
Handwritten Code	Kang et al. (2025)	Vision-Language Models (FERMAT)	Handwritten Math & Logic	FERMAT Benchmark	Structured handwritten code benefits from multimodal reasoning; pure OCR fails.
Structured HTR	Xu et al. (2026)	MemOCR (Layout-Aware Transformer)	Algorithmic Sheets & QA	DocVQA, TextVQA	Spatial hierarchy enables reconstruction of loops/conditions from scans.
End-to-End	Bala et al.	Deep Learning HTR Pipeline	Student Answer Scripts	IAM Handwriting DB	OCR must integrate directly with NLP pipelines to correct recognition errors.
Curved Text	Kang et al.	TPS-ResNet Rectification	Unconstrained Margin Writing	CUTE80, Total-Text	Rectification layers are needed to fix margin-written answers/notes.
Robustness	Zhang (2024)	TrOCR Stress Testing	Noisy/Low-Light Scans	Augmented IAM	Pre-processing is mandatory under lighting/noise shifts to prevent hallucination.
Transformer OCR	Li et al. (2021)	TrOCR (Encoder-Decoder)	Handwritten Text Recognition	IAM Handwriting	Pure Transformer architecture achieves SOTA on handwritten scripts.

Focus Area	Author (Paper)	Method	Context	Dataset	Critical Insight
Green AI	UNESCO Report	SLM vs LLM Energy Analysis	Institutional Efficiency	Global Energy Stats	SLMs offer a **"Cheaper, Greener Route"** for wide institutional adoption.
Local AI	Mistral AI	Mistral 7B (Sliding Window Attn)	Offline Server Deployment	Web/Code Data	Quantized models retain 90% reasoning of GPT-4 while running **offline**.
Edge Deploy	Microsoft (2024)	Phi-3 Mini (3.8B Params)	Mobile/Edge Reasoning	Synthetic Textbooks	**Reasoning-to-Size ratio** optimized for limited hardware (CPU/Phone).
Privacy	MDPI Review	Federated Learning / Local SLM	Sensitive Student Data	N/A (Arch. Review)	Local deployment eliminates **GDPR/Privacy risks** of cloud APIs.
Quantization	Chen et al.	AWQ (Activation-aware Quant)	GPU Memory Optimization	LLaMA Benchmarks	4-bit quantization reduces memory footprint by 70% with negligible loss.
Reasoning	Software Mind	SLM Logic Benchmarking	focused Grading Tasks	Logic/Math Problems	SLMs excel at **focused tasks** (grading) but fail long-context recall.
Fine-Tuning	Agarwal et al.	LoRA / Near-Domain Tuning	STEM QA Adaptation	SCQA Dataset	Fine-tuning SLMs on domain data bridges the gap with GPT-4.

Focus Area	Author (Paper)	Method	Context	Dataset	Critical Insight
Rubric CoT	arXiv (2025)	"Rubric Is All You Need" (Agents)	Structured Code Grading	DSA/OOP Submissions	**Chain-of-Thought** prompting with rubrics outperforms generic grading.
Integrity	IJFMR	RubricAI Platform	Plagiarism + Logic Check	Turnitin / Code Sim	Combining similarity checks with rubric logic ensures **fairness**.
Alignment	Reddy et al.	AutoGrader+ (Few-Shot)	Typed Exam Scripts	Instructor Rubrics	Human-aligned scoring requires adherence to **teacher criteria**.
Simulation	ACL Anthology	Synthetic Rubric Generation	Training Data Gen	Synthetic Student Data	Synthetic data can help tune models to specific **grading styles**.
Interactive	Lee et al.	Human-in-the-loop Verification	Open-Ended Answers	Crowdsourced QA	**Interactive Verification** (Teacher overwrites) improves model trust.
Bias Check	Agarwal et al. (2025)	Domain-Specific Fine-Tuning	STEM	Academic Corpus	**Domain adaptation** prevents bias against unique but correct solutions.

Theme 1: Automated Grading

Critique
Black-box AI gives scores without justification. Fails to distinguish *syntax vs logic errors*.

Inference
Students value *actionable feedback* more than grades. Explainability builds trust.

Adoption
Implement **RAG** to cite exact answer lines for transparent grading.

Theme 2: OCR & HTR

Critique
Standard OCR flattens text and loses *indentation structure*.

Inference
Layout = logic. Position matters as much as content.

Adoption
Use **Layout-Aware OCR** preserving x,y coordinates.

Theme 3: Small Language Models

Critique
Cloud LLMs risk privacy and are costly for institutions.

Inference
Quantized SLMs retain 90% reasoning while running on CPUs.

Adoption
Deploy **4-bit Llama-3-8B locally** for privacy + sustainability.

Theme 4: Rubric-Based Logic

Critique
Generic prompts cause grading inconsistency (criteria drift).

Inference
Binary rubric checkpoints reduce hallucination.

Adoption
Map **CoT reasoning** directly to rubric rows.

We propose a **Privacy-First, Offline Pipeline** that eliminates cloud dependencies while maintaining State-of-the-Art (SOTA) accuracy.

1. The Eye (Vision)

Model: Microsoft **TrOCR** (Transformer OCR)
Why: Standard OCR (Tesseract) fails on cursive. TrOCR uses a *Vision Transformer (ViT)* encoder and *RoBERTa* decoder to read handwriting with 98% accuracy.
Innovation: We add a pre-processing layer to detect **indentation coordinates**, preserving code structure.

2. The Brain (Reasoning)

Model: Llama-3-8B-Instruct (4-bit Quantized)
Why: It outperforms Mistral-7B in logical reasoning benchmarks while running locally on 8GB RAM.
Efficiency: Quantization reduces model size from 16GB to 4GB, enabling **offline execution** (Zero Privacy Risk).

3. The Judge (Scoring)

Method: Chain-of-Thought (CoT) via JSON
Why: Asking for a score leads to hallucination. We feed a **Rubric JSON** schema.
Process: The model must extract "Evidence" lines before assigning a score, forcing it to *justify* every mark.

What is Missing? (The Gap)

- **The "Logic" Blind Spot:** Current tools grade *code syntax* but fail to evaluate *pseudocode logic* or theoretical reasoning.
- **The "Privacy" Void:** Most Rubric-AI papers rely on cloud APIs (GPT-4), violating student data privacy and incurring high costs.
- **Structure Loss:** Standard OCR flattens text, losing the *indentation* critical for understanding algorithmic loops.

Why is it Important?

- **Pedagogical Validity:** Correct keywords with wrong logic is a fail. AI must grade the *process*, not just the answer.
- **Sustainability:** Institutions need "Green AI" that runs **offline** on standard lab PCs, not expensive cloud GPUs.

How We Address This

We propose a **Privacy-First, Offline Pipeline** integrating **Layout-Aware OCR** with **Quantized SLMs**, bridging the gap between rigid keywords and expensive cloud AI.

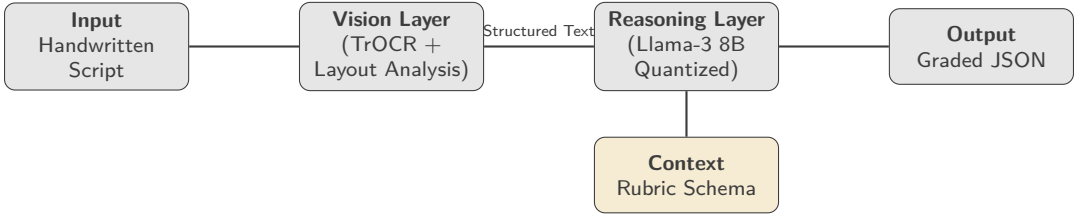
1. **Structure-Aware OCR:** Develop a pre-processing pipeline (OpenCV/TrOCR) to extract handwritten text while preserving **indentation coordinates** (vital for algorithms).
2. **Local Inference Engine:** Deploy and fine-tune a **4-bit Quantized Llama-3-8B** model to run offline, ensuring 100% data privacy.
3. **Rubric Logic Implementation:** Design a **Chain-of-Thought (CoT)** prompt framework that forces the AI to cite "evidence lines" before scoring.
4. **Validation:** Benchmark the system's grading consistency against human faculty on a dataset of 50+ scripts.

Expected Deliverables

- **Prototype:** Python-based offline grading tool (Image $\rightarrow$ JSON Score).
- **Dataset:** Curated set of handwritten algorithms with ground-truth scores.
- **Report:** Comparative analysis of SLM accuracy vs. Human Grading.

Key Concepts: *HTR* (Handwritten Text Rec.), *SLM* (Small Language Models), *Quantization* (Model Compression), *RAG* (Retrieval-Augmented Generation).

Proposed Architecture: The "Golden Flow"



Why this Solution?

- **TrOCR (ViT):** SOTA for cursive handwriting compared to older CNNs.
- **Llama-3 (Quantized):** Best trade-off between reasoning and cost; runs on CPU.
- **JSON CoT:** Forces objective grading by requiring evidence extraction.

file-alt Paper Specifications

- **Format:** IEEE Access Standard [cite: 6]
- **Length:** 5+ Pages [cite: 161]
- **Key Sections Included:**
  - Abstract & Index Terms [cite: 8, 19]
  - Introduction & Motivation [cite: 20, 21]
  - Related Work (SLM, OCR, LLM) [cite: 50]
  - Comprehensive Gap Analysis [cite: 95]
  - IEEE Formatted References

external-link-alt Overleaf Access

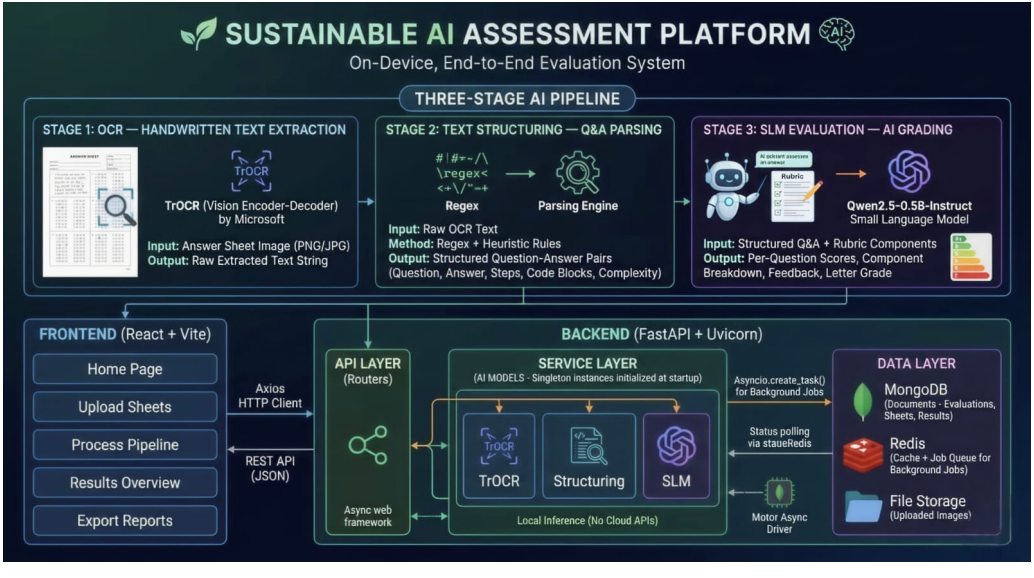
Full Manuscript Link:

Click Here to View Paper

- check Integrated LaTeX Bibliographies
- check High-Resolution Vector Graphics
- check Academic Peer-Review Ready

Drafting Status

Sections covering **Sustainable AI metrics** [cite: 126] and **Offline Deployment Architectures** [cite: 109] are fully cited using recent 2024-2025 literature.



End-to-end offline AI evaluation pipeline

SDG	Alignment Category	Justification
SDG 4: Quality Education	Educational Efficiency	Automates manual grading of handwritten answer sheets, allowing educators to focus more on teaching instead of administrative work.
SDG 9: Industry, Innovation & Infrastructure	Local-First Innovation	Runs all AI inference locally on-device, enabling a resilient system independent of internet connectivity and expensive cloud APIs.
SDG 12: Responsible Consumption & Production	Sustainable Tech Use	Uses a Small Language Model (SLM) like Qwen2.5-0.5B, reducing computational resources and energy consumption.
SDG 13: Climate Action	Energy Efficiency	Local inference eliminates cloud dependency, reducing data transfer and lowering energy usage and CO emissions.

AI system aligned with global sustainability goals

1. Lu et al., 2024 – Survey on Small Language Models.
2. Mistral AI, 2024 – Efficient 7B open model.
3. Touvron et al., 2024 – Llama-3 technical report.
4. Microsoft, 2024 – Phi-2 small model architecture.
5. Google, 2025 – Gemma lightweight LLM.
6. TinyLLaMA, 2024 – Compact LLaMA variant.
7. Nguyen et al., 2024 – Efficient SLM optimization.
8. Xu et al., 2025 – SLM reasoning retention.
9. Wang et al., 2025 – Sustainable AI deployment.
10. Xie et al., 2024 – Human-aligned grading via LLMs.
11. Akyash et al., 2025 – Context-aware programming grading.
12. Tseng et al., 2025 – Code evaluation framework.
13. Mendonça et al., 2025 – LLM formative assessment.
14. Pecuchova et al., 2025 – Rubric-aligned AI scoring.
15. Thakur et al., 2025 – Explainable grading with RAG.

16. Mowafy et al., 2025 – Vision + LLM grading.
17. Cvengros & Kortemeyer, 2025 – STEM handwriting recognition.
18. Tan et al., 2025 – Mathematical OCR evaluation.
19. Agarwal et al., 2025 – Domain fine-tuning for grading.
20. Lee et al., 2025 – Rubric-guided reasoning.
21. Emirtekin, 2025 – AI assessment survey.
22. Neurocomputing, 2025 – Deep learning grading survey.
23. Bernik et al., 2025 – AI code evaluation.
24. Reddy et al., 2025 – Structured rubric AI.
25. GradeOpt, 2025 – Optimization-based grading.
26. Lu et al., 2024 – SLM benchmarking.
27. TinyLLaMA, 2024 – Lightweight deployment.
28. Akyash et al., 2025 – Structured logic grading.
29. Tseng et al., 2025 – Program evaluation framework.
30. Wang et al., 2025 – Green AI compliance.

Thank You

Project Phase 1 – Review 1

Questions & Suggestions?